

Chang Zeng

Senior Research Scientist | Generative Multimodal AI | Audio-Visual AI

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| github.com/zengchang233 | scholar.google.com/citations?user=gfGyn49j-MkC&hl=en

Senior Research Scientist in **Multimodal LLMs** with **7+ years** of experience from research to production. I have led end-to-end development of **expressive Multimodal** systems and served as a main contributor to a **full-duplex speech** system, covering multimodal data curation, audio tokenization/codec design, large-scale multi-GPU training, and deployment. My work has delivered state-of-the-art quality-efficiency trade-offs (including sub-1 kbps codec performance) and top benchmark rankings in multilingual settings. I publish actively at ICLR/ICML/Interspeech and specialize in turning advanced methods (**GANs, diffusion, transformers**) into **product-ready Multimodal AI**.

Education

National Institute of Informatics (SOKENDAI) 2020.10 - 2024.03.22

Doctor of Informatics

Sponsored by the Japanese government MEXT scholarship, supervised by Prof. [Yamagishi](#).

The University of Tokyo 2017.10 - 2020.03

Master of Electrical Engineering and Information System

Supervised by Prof. [Minematsu](#).

Tianjin University 2012.09 - 2016.07

Bachelor of Engineering

Publications

Featured Publications

- Zeng, C.**, & Wang, C. (2024). InstructSing: High-fidelity Singing Voice Generation via Instructing Yourself. *IEEE SLT 2024*. [Demo page](#).
- Zeng, C.**, Miao, X., Wang, X., Cooper, E., & Yamagishi, J. (2024). [Joint Speaker Encoder and Neural Back-end Model for Fully End-to-End Automatic Speaker Verification with Multiple Enrollment Utterances](#). *Journal of Computer Speech & Language*. (JCR Q1)
- Zeng, C.**, Wang, X., Cooper, E., Miao, X., & Yamagishi, J. (2022). [Attention back-end for automatic speaker verification with multiple enrollment utterances](#). In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 6717-6721). [Project page](#). (CoreRank B)
- Wang, C., **Zeng, C (co-first author)**., & He, X. (2023). [Xiaoicesing 2: A High-Fidelity Singing Voice Synthesizer Based on Generative Adversarial Network](#). In *Proc. Interspeech 2023*, 5401-5405. [Project page](#). (CoreRank A)
- Wang, C., **Zeng, C (Co-first author)**., Zhang, B., Ma, Z., Zhu, Y., Cai, Z., Zhao, J., Jiang, Z., & Chen, Y. (2024). [HAM-TTS: Hierarchical Acoustic Modeling for Token-Based Zero-Shot Text-to-Speech with Model and Data Scaling](#). [Demo Page](#).
- Li, K., Sang, W., **Zeng, C.**, Yang, R., Chen, G., Hu, X. (2024). [LibriSpace: A Simulated Moving Audio Dataset for Speech Enhancement and Separation](#). *ICLR 2025*.

Full list: Google Scholar: <https://scholar.google.com/citations?user=gfGyn49j-MkC&hl=en>

Work Experiences

Shanda AI Research Tokyo 2025.09 - Present

Senior AI Researcher

Responsibilities

- Lead R&D of unified audio-visual generative models and full-stack interactive intelligence systems for digital humans and multimodal gaming environments.

Achievements

- Led a team of **2 full-time researchers + 1 intern** to build an end-to-end multimodal omni-avatar system from scratch on a Qwen-based foundation model, seamlessly coupling expressive voice generation (Talker) with synchronized facial animation generation (Animator).
- Built a robust pipeline covering large-scale **audio-video paired data curation, multi-modal codec design, and multi-node multi-GPU LLM training**. Achieved SOTA tokenization efficiency (sub-1 kbps for audio) while ensuring extremely low latency for real-time interactive intelligence.
- Reached **#2 ranking** across 3 languages and 2 evaluation metrics against major Chinese tech competitors, specifically optimizing for personality-aligned, cross-modal consistency in **zero-shot text-to-audio-visual generation**.
- Serve as Co-PI in a joint project with Tsinghua University on complex **audio-visual scene understanding** (e.g., cocktail-party interaction). Delivered an LLM-based universal multimodal data pipeline (HIVE) and model architecture, aiming at exploring visual cues and multi-sensory alignment for robust audio separation.

Li Auto 2024.04 - 2025.09

Multimodal Generative Researcher

Responsibilities

- Developed voice generation systems for Li Auto smart-space products and contributed to the multimodal foundation model **MindGPT-4o**.

Achievements

- Proposed **GFSQ (Group Finite Scalar Vector Quantization)** tokenizer for GPT-SoVITS, improving codebook utilization and decoding quality via delay-pattern decoding.

- Trained a **multi-timbre, multi-style** voice generation model for in-car voice-blog scenarios, enabling high-quality style/voice selection in production use cases.
- Built synthetic-data workflows with internal foundation TTS models to scale data diversity (accent, dialect, language, emotion, scenario).
- Led data production and **audio-head pre-/post-training** for a full-duplex conversational model, improving customization capability and robustness of speech interaction.

Xiaoice

2022.07 - 2023.07

Avatar Research Intern

Responsibilities

- Focused on high-fidelity (48kHz) singing voice generation in collaboration with research and engineering teams.

Achievements

- Upgraded FastSpeech2-based XiaoiceSing to **XiaoiceSing2** using adversarial training; achieved near-human MOS (**4.23 vs 4.27**).
- Developed **HiFi-WaveGAN** vocoder with a novel pulse-sequence design, outperforming HiFi-GAN on 48kHz singing synthesis MOS (**4.23 vs 3.89**).
- Built **CrossSinger** (cross-lingual, multi-singer SVS in EN/JP/CN) with language-aware encoding and a bias eliminator; accepted at **ASRU 2023**.
- Improved training efficiency with **InstructSing** (DDSP + adversarial + UNet refinement), reducing convergence to **~20k steps** (about **10× faster** than common neural vocoders).
- Explored hierarchical acoustic modeling for token-based voice LMs (e.g., VALL-E style), improving pronunciation consistency.

ALIBABA

2020.04 - 2020.09

Speech Recognition Researcher (Full-time)

Responsibilities

- Developed speech AI systems for Taobao Live compliance and broadcaster-risk control.

Achievements

- Built a large-scale **speaker recognition** system to verify broadcaster identity at scale in livestream scenarios.
- Developed a **spoken-term detection** pipeline for policy-sensitive/illegal word monitoring in live broadcasts.
- Researched SSL-based speech representations and implemented an end-to-end ASR system with knowledge distillation (ESPnet-based) for improved accuracy-efficiency tradeoff.

Activities

Competitions

- 4th/77 place for VoxCeleb Speaker Recognition Challenge 2019.
- 2nd/110 place for Zhijiang Cup Speech Recognition for Conversational Scenario 2021.
- 4th/42 place for Audio Deep Synthesis Detection Challenge 2022 track1 (Low-quality Fake Audio Detection, LF).
- 5th/27 place for Audio Deep Synthesis Detection Challenge 2022 track2 (Partially Fake Audio Detection, PF).

Academic Activities

- Research assistant at Yamagishi Lab, National Institute of Informatics
- ICASSP 2022 Oral Presentation.
- Interspeech 2022 Oral Presentation.
- Interspeech 2023 Poster Presentation.
- Reviewer of [IEEE Open Journal of Signal Processing](#).
- Reviewer of ICASSP, ICME, Interspeech, NeurIPS, ICLR, ICML, and AISTATS.
- Organization committee of [Joint Workshop of VoicePersonae and ASVspoof 2023](#).

Open Source

- [WeSpeaker](#) contributor (1.2k stars), [ASV-Subtools](#) contributor (636 stars), [HIVE dataset](#) (227 stars, 10k downloads on HF)

Skills

Tools and Languages	Python, CPP, Shell, Git, MySQL
Toolkits	PyTorch, Pytorch-Lightning, SpeechBrain, WeNet, WeSpeaker, Kaldi, Espnet, HuggingFace
Communication	Chinese (native), English (business level), Japanese (N2-125)